

Trial Higher School Certificate Examination

2006



Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Write your student number on every booklet
- Begin each question in a new booklet
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks (120)

- Attempt Questions 1-10
- All questions are of equal value

Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

Question 1 (12 marks) – Start a new booklet

Marks

- a) Find the value of $\frac{16.2}{84.7 \times 16.8 + \sqrt{504.3}}$ correct to 3 significant figures 2
- b) Write $\frac{\sqrt{3}+1}{\sqrt{3}-1}$ in the form $a + b\sqrt{3}$ 2
- c) Simplify $\frac{2x}{3} - \frac{x+2}{5}$ 2
- d) Solve $|5-2x| \geq 3$ 2
- e) Factorise $16a^4 - 1$ 2
- f) Find the primitive of $x^2\sqrt{x}$ 2

Question 2 (12 marks) – Start a new booklet

Marks

- a) The points A(2, -2), B(-2, -3) and C(0, 2) are the vertices of a triangle ABC
- (i) Plot these points on the number plane 1
- (ii) Find the gradient of AC 1
- (iii) Find the angle of inclination of AC to the positive x -axis, to the nearest degree 1
- (iv) Show that the equation of AC is $2x + y - 2 = 0$ 1
- (v) Calculate the perpendicular distance of B from the side AC 1
- (vi) Hence find the area of $\triangle ABC$ 2
- (vii) Find the co-ordinates of D such that ABCD is a parallelogram 1
- b) In $\triangle ABC$, $AB = 2\text{cm}$, $\angle ABC = 105^\circ$ and $\angle BCA = 30^\circ$. Find the length of BC 2
- c) For what values of k will $x^2 + 3x + k$ be positive definite? 2

Question 3 (12 marks) – Start a new booklet

Marks

- a) Differentiate with respect to x :
- (i) $(e^x + e^{-x})^3$ 2
- (ii) $x^2 \sin 3x$ 2
- b) (i) Find $\int \frac{3}{5x-3} dx$ 2
- (ii) Evaluate $\int_0^{\pi} 4 \sec^2 x dx$ 2
- c) A ship sails from a Port A 50 nautical miles due east to a Port B. It then proceeds a distance of 20 nautical miles on a bearing of 020°T to a Port C.
- (i) Find the distance of Port C from Port A (correct to 2 decimal places) 2
- (ii) Find the bearing of Port C from Port A 2

Question 4 (12 marks) – Start a new booklet

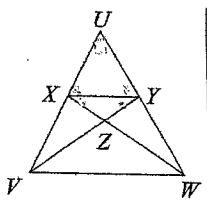
Marks

- a) A circular sector AOB, centre O and radius 8cm, contains an angle of $\frac{3\pi}{4}$ radians at the centre. The straight edges OA and OB are joined to form a right circular cone.

Find the:

- (i) exact length of the minor arc AB 1
- (ii) radius of the base of the cone 2
- (iii) curved surface area of the cone 1
- b) Explain why the curve $y = \frac{3x+5}{x+1}$ is always decreasing for all real values of x except $x = -1$ 3
- c) Evaluate $\sum_{n=3}^5 \frac{(-1)^n}{2n+1}$ 1

- d) In the diagram below $\angle UXY = \angle UYX$ and $XZ = YZ$



Copy the diagram into your answer booklet.
Prove that $\triangle UXY \cong \triangle UWX$

4

Question 5 (12 marks) – Start a new booklet

Marks

- a) (i) Copy and complete the table of values for $y = \frac{1}{x+2}$ in your writing booklet 1

x	0	0.5	1	1.5	2
y					

- (ii) By using Simpson's rule with 5 function values, estimate the value of the integral $\int_0^2 \frac{1}{x+2} dx$ 3

- b) The rate at which liquid is flowing into a vessel after t minutes is given by $\frac{dV}{dt} = \frac{1}{t+1}$
- (i) If $(\log_e 2) \text{ m}^3$ of liquid flows into the vessel after 3 minutes, find an expression for $V(t)$, the volume of water in the tank at time t minutes. 2
- (ii) Find the volume of the liquid in the vessel after 8 minutes. Give your answer in exact simplest form 2
- c) During the first week in January a car dealer sold 15 cars. In the second week he sold 18 cars, and in each succeeding week after that he sold 3 more cars than he sold the previous week.
- (i) How many cars did he sell during the last week in December? 2
- (ii) Calculate the total number of cars he sold during the year. 2

Question 6 (12 marks) – Start a new booklet

Marks

- a) (i) For what values of x will a limiting sum exist for the geometric series $1 - 3x + 9x^2 - \dots$? 2
- (ii) Find the value of x for which the limiting sum is $\frac{4}{5}$ 2
- b) The price of one tonne of copper, \$P, was studied over a period of t years.
- (i) Throughout the period of study $\frac{dP}{dt} > 0$.
What does this say about the price of copper? 1
- (ii) It was also observed that the rate of change of the price of copper decreased over the period of study. What does this statement imply about $\frac{d^2P}{dt^2}$? 1
- c) (i) Sketch the parabola which has focus (1, -5) and directrix $y = 1$, showing the co-ordinates of the vertex. 3
- (ii) Write down the equation of the parabola 1
- d) Find the equation of the tangent to the curve $y = e^{2x-3}$ at the point where $x = \frac{3}{2}$ 2

Question 7 (12 marks) – Start a new booklet

Marks

- a) (i) Find the co-ordinates of the points where the curves $y = \sqrt{x}$ and $y = x^2$ intersect 2
- (ii) Sketch $y = \sqrt{x}$ and $y = x^2$ on the same axes 2
- (iii) Find the volume of the solid generated by rotating the region enclosed between the curves $y = \sqrt{x}$ and $y = x^2$ about the y -axis 3
- b) Sue contributes \$1 200 every year, for 20 years, to a superannuation account which pays 9%p.a. compounded monthly.
- (i) Show that her total investment at the end of 2 years can be represented by : 2
- $$1\,200(1.0075)^{24} + 1\,200(1.0075)^{12}$$
- (ii) Hence, find the value of her investment at the end of 20 years. 3

Question 8 (12 marks) – Start a new booklet

Mark

- a) The population P of a town at the end of t years is given by $P = Ae^{kt}$, where A and k are constants. After 1 year the population is 1 020.

(i) Find the value of A if the initial population was 1 000

1

(ii) Find the value of k

2

(iii) Calculate the population after 10 years

1

(iv) How many years will it take the population of the town to double?

2

- b) (i) Show that $x = \frac{\pi}{8}$ is a solution to $y = 3\sin 2x$ and $y = 3\cos 2x$

1

(ii) On the same axes sketch $y = 3\sin 2x$ and $y = 3\cos 2x$ for $0 \leq x \leq \pi$

2

(iii) Find the area bounded by the curves $y = 3\sin 2x$, $y = 3\cos 2x$ and the x -axis in the domain $0 \leq x \leq \frac{\pi}{4}$

3

Question 9 (12 marks) – Start a new booklet

Mark

- a) Joe deposited \$20 000 at the beginning of January into an account which paid interest at the rate of $\frac{1}{2}\%$ per month compounded monthly. He withdrew \$50 each month from the account immediately after the interest was paid.

(i) How much money did Joe have in the account after making the first withdrawal?

1

(ii) Show that after making the n th withdrawal, his balance in the account is given by the expression:

3

$$10\,000 \times 1.005^n + 10\,000$$

(iii) Find the minimum number of withdrawals needed for his account balance to show at least \$50 000

2

b) (i) State the condition necessary for a quadratic equation to have rational roots

1

(ii) Prove that the equation $3px^2 = 2px + 3qx - 2q$, where p and q are rational, has rational roots for all values of p and q

4

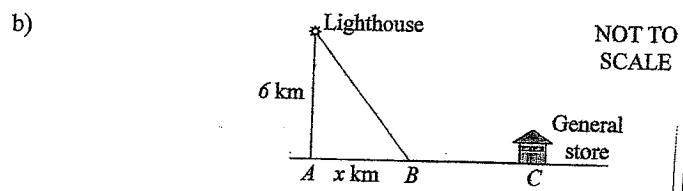
(iii) What can be concluded about the number of the roots if $p = \frac{3q}{2}$?

1

Question 10 (12 marks) – Start a new booklet

Marks

- a) (i) Given that the curve $y = x^2 e^x$ has turning points at $(0, 0)$ and $(-2, \frac{4}{e^2})$, determine their nature using the first derivative 3
- (ii) Discuss the behaviour of the curve for large positive and large negative values of x 1
- (iii) Sketch the curve $y = x^2 e^x$ 1



The water's edge is a straight line ABC which runs east-west. A lighthouse is 6km due north of A.

10km due east of A is the general store. To get to the general store as quickly as possible, the lighthouse keeper rows to a point B, x km from A, and then jogs to the general store. The lighthouse keeper's rowing speed is 6km/h and his jogging speed is 10km/h.

- (i) Show that it takes the lighthouse keeper $\frac{\sqrt{36+x^2}}{6}$ hours to row from the lighthouse to B 2
- (ii) Show that the total time taken for the lighthouse keeper to reach the general store is given by 1
- $$T = \frac{\sqrt{36+x^2}}{6} + \frac{10-x}{10} \text{ hours}$$
- (iii) Hence, show that when $x = 4\frac{1}{2}$ km the time it takes for the lighthouse keeper to travel from the lighthouse to the general store is a minimum 3
- (iv) Hence find the quickest time it takes the lighthouse keeper to go to the general store from the lighthouse. Give your answer correct to the nearest minute. 1

End of Paper

MATHEMATICS TRIAL SOLUTIONS 2006.

i) $0.011207 \dots$
 $= 0.0112$ (to 3 sig. figs)

ii) $\frac{\sqrt{3}+1}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} = \frac{(\sqrt{3}+1)^2}{3-1}$
 $= \frac{3+2\sqrt{3}+1}{2}$
 $= \frac{4+2\sqrt{3}}{2}$
 $= 2+\sqrt{3}$

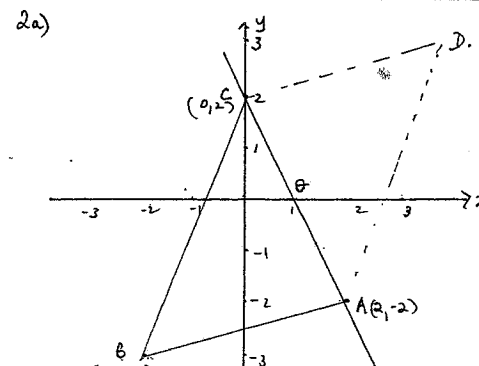
iii) $\frac{2x}{3} - \frac{x+2}{5} = \frac{10x}{15} - \frac{3(x+2)}{15}$
 $= \frac{10x-3x-6}{15}$
 $= \frac{7x-6}{15}$

iv) $|5-2x| \geq 3$

$5-2x \geq 3$ $5-2x \leq -3$
 $2 \geq 2x$ $8 \leq 2x$
 $x \leq 1$ $4 \leq x$
 $\therefore x \geq 4$

v) $16a^4 - 1 = (4a^2 - 1)(4a^2 + 1)$
 $= (2a - 1)(2a + 1)(4a^2 + 1)$

vi) $\int x^2 \sqrt{x} dx = \int x^{5/2} dx$
 $= \frac{2}{7} x^{7/2} + C$



ii) $m_{AC} = \frac{2-2}{0-2}$
 $= -2$

iii) let θ be angle of inclination
 $\therefore \tan \theta = -2$
 $\therefore \theta = 110^\circ$ (to next degree)

iv) $y = mx + b$
 $y = -2x + 2$
 $\therefore 2x + y - 2 = 0$

v) $d = \frac{|-2 \times 2 - 3 \times 1 - 2|}{\sqrt{2^2 + 1^2}}$
 $= \frac{9}{\sqrt{5}}$

vi) Area of $\triangle ABC = \frac{1}{2} \times AC \times \frac{10}{\sqrt{5}}$
length $AC = \sqrt{(2+2)^2 + (0-2)^2}$
 $= \sqrt{20}$
 $\therefore \text{area} = \frac{1}{2} \times \sqrt{20} \times \frac{10}{\sqrt{5}}$
 $= 9 \text{ sq. units.}$

vii) midpt AC = $(1, 0)$
 $\therefore$ midpt BD = $(1, 0)$
 $\therefore D = (1+3, 0+3)$
 $= (4, 3)$

b)

$\frac{BC}{\sin 45} = \frac{2}{\sin 30}$
 $\therefore BC = \frac{2 \sin 45}{\sin 30}$
 $= \frac{2 \times \frac{1}{\sqrt{2}}}{\frac{1}{2}}$
 $= \frac{4}{\sqrt{2}} \approx 2.83$ (to 2dp)

c) $x^2 + 3x + k$
For positive definite
 $a > 0$ and $\Delta < 0$
 $a = 1$ $b^2 - 4ac < 0$
 $\therefore a > 0$ $9 - 4 \times 1 \times k < 0$
 $9 < 4k$

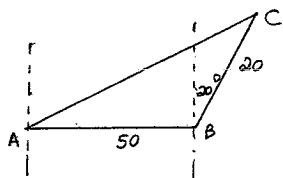
3a) i) $y = (e^x + e^{-x})^3$
 $y' = 3(e^x + e^{-x})^2 (e^x - e^{-x})$

ii) $y = x^2 \sin 3x$
 $y' = (\sin 3x) 2x + x^2 3 \cos 3x$
 $= 2x \sin 3x + 3x^2 \cos 3x$

b) i) $\int \frac{3}{5x-3} dx = \frac{3}{5} \ln(5x-3) + c$

ii) $\int_0^{\frac{\pi}{4}} \sec^2 x dx = [\tan x]_0^{\frac{\pi}{4}}$
 $= \tan \frac{\pi}{4} - \tan 0$
 $= 1$

c)



i) $AC^2 = 50^2 + 20^2 - 2 \times 50 \times 20 \times \cos 110$
 $AC = 59.87 \text{ nm. (to 2dp)}$

ii) $\frac{20}{\sin \hat{CAB}} = \frac{59.87}{\sin 110}$

$\therefore \sin \hat{CAB} = \frac{20 \sin 110}{59.87}$

$\hat{CAB} = 18^\circ 18'$

$\therefore \text{Bearing of C from A} = 90 - 18^\circ 18'$
 $= 71^\circ 42'$

4.a) i) $l = 8 \times \frac{3\pi}{4}$

$\therefore \text{arc AB} = 6\pi$

ii) $2\pi r = 6\pi$
 $\therefore r = 3$

iii) C.S.A. = $\pi r l$
 $= \pi \times 3 \times 8$

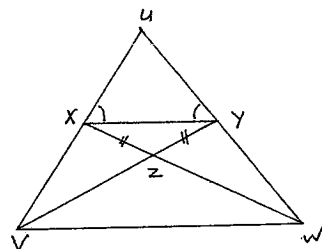
b) $y = \frac{3x+5}{x+1}$
 $y' = \frac{(x+1) \times 3 - (3x+5) \times 1}{(x+1)^2}$
 $= \frac{-2}{(x+1)^2}$

If curve is decreasing $y' < 0$
 Since $(x+1)^2 > 0$ for all x except $x = -1$
 and $-2 < 0$

$\therefore \frac{-2}{(x+1)^2} < 0$ for all x except $x = -1$
 $\therefore$ curve is decreasing for all x except $x = -1$.

c) $\sum_{n=3}^5 \frac{(-1)^n}{2n+1} = \frac{-1}{7} + \frac{1}{9} - \frac{1}{11}$
 $= \frac{-85}{693}$

d)



$XU = YU$ (In triangle, sides opposite equal angles are equal)

$\hat{ZXY} = \hat{ZYX}$ (In triangle, angles opposite equal sides are equal)

$\hat{UXY} = \hat{UYX}$ (given)

$\therefore \hat{UXZ} = \hat{UYZ}$ Since $\hat{UXW} = \hat{UXY} + \hat{YXZ}$
 and $\hat{UYV} = \hat{UYX} + \hat{XYZ}$

In $\triangle VYZ, \triangle UWZ$

1. $\hat{VUW}$ is common

2. $\hat{UXW} = \hat{UYV}$ (proven above)

3. $XU = YU$ (proven above)

$\therefore \triangle UVY \equiv \triangle UWZ$ (by A.A.S. test)

5) i)

x	0	$\frac{1}{2}$	1	$\frac{1}{2}$	2
y	$\frac{1}{2}$	$\frac{2}{5}$	$\frac{1}{3}$	$\frac{2}{7}$	$\frac{1}{4}$

$y = \frac{1}{x+2}$

ii) $\int_0^2 \frac{1}{x+2} dx$
 $= \frac{1-0}{6} \left[\frac{1}{2} + \frac{2}{5} \times 4 + \frac{1}{3} \right] + \frac{2-1}{6} \left[\frac{1}{3} + 4 \times \frac{2}{7} + \frac{1}{4} \right]$
 $= \frac{1747}{2520}$
 $= 0.693$ (to 3 dp)

b) $\frac{dV}{dt} = \frac{1}{t+1}$

i) $V = \int \frac{1}{t+1} dt$

$V = \ln(t+1) + c$

$t=3, V = \ln 2: \ln 2 = \ln 4 + c$

$\ln \frac{1}{2} = c$

$\therefore V = \ln(t+1) + \ln \frac{1}{2}$

ii) $t=8: V = \ln 9 + \ln \frac{1}{2}$

$= \ln(9 \times \frac{1}{2})$

$= \ln 4.5$

c) Sequence 15, 18, 21, ...
 Arithmetic where $a = 15, d = 3$

i) $T_{52} = a + 51d$

$= 15 + 51 \times 3$

$= 168$

$\therefore$ 168 cars sold during 1st week of December.

ii) $S_{52} = \frac{52}{2} (15 + 168)$

$= 4758$

$\therefore$ Total number of cars sold for year is 4758

6.a) i) For S_{∞} to exist $-1 < r < 1$

$1 - 3x + 9x^2 \dots$

$r = -3x$

$\therefore -1 < -3x < 1$

$\frac{1}{3} > x > -\frac{1}{3}$

i.e. $-\frac{1}{3} < x < \frac{1}{3}$

ii) $\frac{4}{5} = \frac{1}{1+3x} \quad S_{\infty} = \frac{a}{1-r}$

$4 + 12x = 5$

$12x = 1$

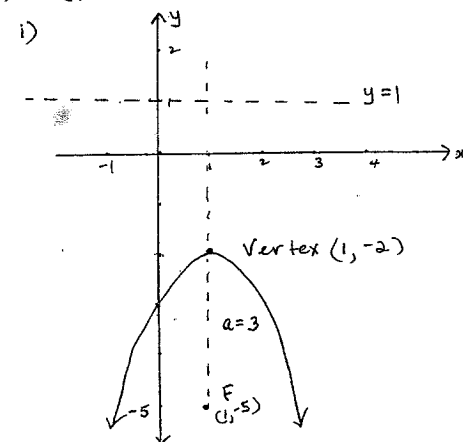
$x = \frac{1}{12}$

b) i) price of copper was rising

ii) $\frac{d^2P}{dt^2} < 0$

c) F(1, -5) D: $y = 1$

i)



ii) $4x(y+2) = (x-1)^2$

$a = 3 \quad \therefore (x-1)^2 = -12(y+2)$

d) $y = e^{2x-3}$

$y' = 2e^{2x-3}$

at $x = \frac{3}{2}: m = 2e^0$
 $= 2$

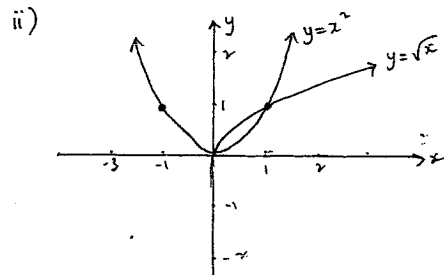
$x = \frac{3}{2}: y = e^0 = 1$

$\therefore$ Eqn of tangent: $y - 1 = 2(x - \frac{3}{2})$

$y - 1 = 2x - 3$

$y = 2x - 2$

7. a) i) $\sqrt{x} = x^2$
 $x = x^4$
 $x^4 - x = 0$
 $x^3(x-1) = 0$
 $x=0$ or $x=1$
 $y=0$ $y=1$



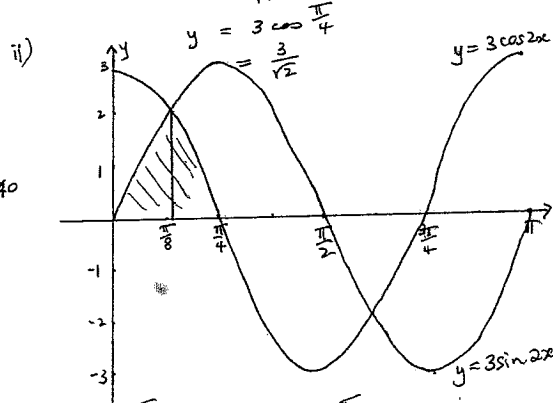
iii) $y = \sqrt{x}$
 $y^2 = x$
 $y^4 = x^2$
 $\text{Vol}_y = \pi \int_0^1 y - y^4 dy$
 $= \pi \left[\frac{y^2}{2} - \frac{y^5}{5} \right]_0^1$
 $= \pi \left[\left(\frac{1}{2} - \frac{1}{5} \right) - 0 \right]$
 $= \frac{3\pi}{10} \text{ cu. units}$

b) i) 9% pa = 0.0075 per month
 Value of first \$1200 is $1200(1.0075)^{24}$
 Value of second \$1200 is $1200(1.0075)^{12}$
 $\therefore$ Total value at end of 2 years = $1200(1.0075)^{24} + 1200(1.0075)^{12}$
 ii) Total value after 20 yrs = $1200(1.0075)^{12} + \dots + 1200(1.0075)^{240}$
 $= 1200(1.0075^{12} + \dots + 1.0075^{240})$
 Sum of geometric series
 where $a = 1.0075^{12}$
 $r = 1.0075^{12}$
 $n = 20$
 $\therefore \text{Value} = 1200 \times \frac{1.0075^{12}(1.0075^{240} - 1)}{1.0075^{12} - 1}$
 $\approx \$70,089.23$

8a) $P = Ae^{kt}$
 i) $t=0$, $P = 1000$
 $\therefore 1000 = Ae^0$
 $\therefore A = 1000$
 ii) $P = 1000e^{kt}$
 $t=1$, $P = 1020$: $1020 = 1000e^k$
 $\frac{1020}{1000} = e^k$
 $\ln \frac{1020}{1000} = k$
 $\therefore k = 0.0198$ (to 4 d.p.)
 iii) $t=10$: $P = 1000e^{10k}$
 $= 1000e^{0.198}$
 $P = 669.967$
 $\therefore$ population is 670 (to next person)
 iv) $2000 = 1000e^{kt}$
 $2 = e^{kt}$
 $\frac{\ln 2}{k} = t$

$\therefore t = 35.007$
 $\therefore$ town's population doubles in about 35 yrs.

b) i) $x = \frac{\pi}{8}$: $y = 3 \sin \frac{\pi}{4}$
 $= \frac{3}{\sqrt{2}}$



iii) Area = $\int_{\pi/8}^{5\pi/8} 3 \sin 2x dx + \int_{\pi/8}^{5\pi/8} 3 \cos 2x dx$
 $= -\frac{3}{2} [\cos 2x]_{\pi/8}^{5\pi/8} + \frac{3}{2} [\sin 2x]_{\pi/8}^{5\pi/8}$
 $= -\frac{3}{2} \left(\frac{1}{\sqrt{2}} - 1 \right) + \frac{3}{2} \left[1 - \frac{1}{\sqrt{2}} \right]$

9. i) Amount = $20000(1.005) - 50$
 $= 20050$

ii) Let A_n be the amount in the account after the n th withdrawal
 $A_1 = 20000(1.005) - 50$
 $A_2 = [20000(1.005) - 50] \times 1.005 - 50$
 $= 20000(1.005)^2 - 50 \times 1.005 - 50$
 $= 20000(1.005)^2 - 50(1 + 1.005)$

Similarly
 $A_n = 20000(1.005)^n - 50(1 + 1.005 + \dots + 1.005^{n-1})$
 $= 20000(1.005)^n - 50 \times \text{sum of geom series.}$

Sum = $\frac{a(r^n - 1)}{r - 1}$
 $= \frac{1 \times (1.005^n - 1)}{1.005 - 1}$
 $= \frac{1.005^n - 1}{0.005}$
 $= 200(1.005^n - 1)$

$\therefore A_n = 20000(1.005)^n - 50 \times 200(1.005^n - 1)$
 $= 20000(1.005)^n - 10000(1.005^n - 1)$
 $= 20000(1.005^n) - 10000 \times 1.005^n + 10000$
 $= 10000(1.005^n) + 10000$

iii) If $A_n = 50,000$ $n = ?$
 $50,000 = 10,000(1.005^n) + 10,000$
 $\frac{40,000}{10,000} = 1.005^n$
 $4 = 1.005^n$
 $\ln 4 = n \ln 1.005$
 $n = \frac{\ln 4}{\ln 1.005}$
 $n = 277.95 \dots$

$\therefore$ min. number of withdrawals is 278.

b) i) $b^2 - 4ac > 0$ and $b^2 - 4ac$ is a perfect square.

ii) $3px^2 = 2px + 3q$
 $3px^2 - (2p + 3q)x + 2q = 0$
 $b^2 - 4ac = [-(2p + 3q)]^2 - 4 \times 3p \times 2q$
 $= 4p^2 + 12pq + 9q^2 - 24pq$
 $= 4p^2 - 12pq + 9q^2$
 $= (2p - 3q)^2$

Since $b^2 - 4ac > 0$ and a perfect square, roots will be rational if p and q are rational.

(iii) If $p = \frac{3q}{2}$
 then $b^2 - 4ac = 0$
 $\therefore$ there is only one rational root.

10a) i) $y = x^2 e^x$

$$y' = e^x 2x + x^2 e^x$$

$$= x e^x (2 + x)$$

For point (0,0)

$$x = -\frac{1}{2} : y' = -\frac{1}{2} e^{-\frac{1}{2}} \left(2 - \frac{1}{2}\right)$$

$$x = \frac{1}{2} : \quad \quad \quad \doteq -0.45 \dots$$

$$y' = \frac{1}{2} e^{\frac{1}{2}} \left(2 + \frac{1}{2}\right)$$

$$\doteq 2.06 \dots$$

 $\therefore (0,0)$ is a minimum turning point

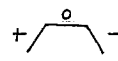
For $\left(-2, \frac{4}{e^2}\right)$

$$x = -2\frac{1}{2} : y' = -2 e^{-2} \left(2 - 2\frac{1}{2}\right)$$

$$\doteq 0.135 \dots$$

$$x = -1\frac{1}{2} : y' = -1\frac{1}{2} e^{-1\frac{1}{2}} \left(2 - 1\frac{1}{2}\right)$$

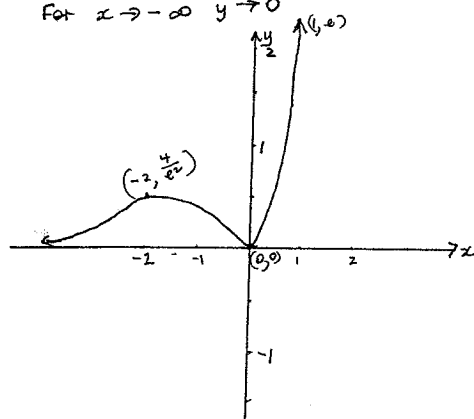
$$\doteq -0.167$$

 $\therefore \left(-2, \frac{4}{e^2}\right)$ is a maximum turning point.

ii) $y = x^2 e^x$

For $x \rightarrow \infty$ $y \rightarrow \infty$

For $x \rightarrow -\infty$ $y \rightarrow 0$



b) i) Distance from lighthouse to B = $\sqrt{6^2 + x^2}$

$$= \sqrt{36 + x^2}$$

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$\therefore \text{Time from Lighthouse to B} = \frac{\sqrt{36+x^2}}{6}$$

ii) Distance from B to C = $10 - x$

$$\therefore \text{time} = \frac{10-x}{10}$$

$$\therefore \text{Total time: } T = \frac{\sqrt{36+x^2}}{6} + \frac{10-x}{10}$$

iii) For minimum $\frac{dT}{dx} = 0$ and $\frac{dT}{dx^2} > 0$

$$T = \frac{\sqrt{36+x^2}}{6} + \frac{10-x}{10}$$

$$\frac{dT}{dx} = \frac{\frac{1}{2} (36+x^2)^{-1/2} \times 2x}{6} - \frac{1}{10}$$

$$= \frac{x}{6\sqrt{36+x^2}} - \frac{1}{10}$$

For minimum let $\frac{dT}{dx} = 0$.

$$\frac{x}{6\sqrt{36+x^2}} = \frac{1}{10}$$

$$10x = 6\sqrt{36+x^2}$$

$$100x^2 = 36(36+x^2)$$

$$100x^2 = 1296 + 36x^2$$

$$64x^2 = 1296$$

$$x^2 = 20\frac{1}{4}$$

$$x = \pm 4.5$$

Since x represents distance $x > 0$

$$\therefore x = 4.5$$

To prove a minimum use sign change in $\frac{dT}{dx}$

$$x = 4 : \frac{dT}{dx} = \frac{4}{6\sqrt{52}} - \frac{1}{10}$$

$$\doteq -7.5$$

$$x = 5 : \frac{dT}{dx} = \frac{5}{6\sqrt{61}} - \frac{1}{10}$$

$$\doteq 6.7$$

Since $-\frac{dT}{dx} < 0$ $\therefore x = 4.5$ will give minimum time

iv) $x = 4\frac{1}{2} : T = \frac{\sqrt{36+4.5^2}}{6} + \frac{10-4\frac{1}{2}}{10}$